Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

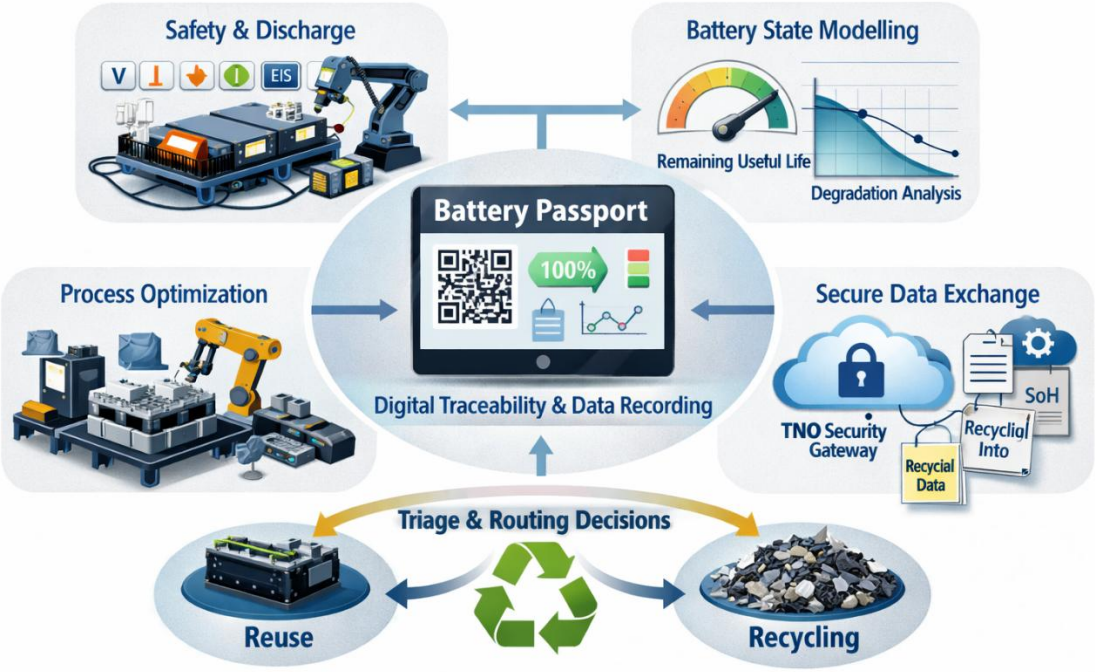
1. Technology Proposer Information

Proposer	Name	Steven Wilkins	Email	steven.wilkins@tno.nl
Institute Name	TNO (Netherlands Organisation for Applied Scientific Research)			
Address	Automotive Campus 30, 5708 JZ Helmond, The Netherlands Postal address: P.O. Box 756, 5700 AT Helmond, The Netherlands			

2. Technology Overview

Title of Proposed Technology	Secure and traceable pre-treatment for closed-loop EV battery recycling, with EU-compliant battery passport integration			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery (√), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical & Electronics		Machinery & Materials	
	Medium Category	Small Category	Medium Category	Small Category
	Batteries	Battery materials	Clean Production	Resource Recycling Technology
Concept of Proposed Technology	This work focuses on the recycling pre-treatment stage for end-of-life EV batteries, covering controlled discharge, dismantling, shredding and separation, and adds a digital traceability layer that records process and battery-condition data in a battery passport, which the EU Battery Regulation requires (from 18 February 2027) for EV batteries and industrial batteries above 2 kWh and makes accessible via a QR code. The proposed stack starts with a safety and efficiency			

module for risk-aware triage and controlled discharge using electrical measurements (voltage and current), temperature sensing, and off-gas detection for early-warning signals, with optional Electrochemical Impedance Spectroscopy and Hardware-in-the-Loop validation leveraging TNO's Battery Lab capabilities.



A battery state modelling layer provides remaining useful life and degradation estimates, building on peer-reviewed work, and feeds decision logic for routing batteries toward reuse, repurposing, remanufacturing, or recycling, consistent with the regulation's intent that passport information supports choosing between these pathways. Process optimization then configures dismantling and shredding parameters to maximize black mass recovery efficiency and reduce losses of critical metals to coarse fractions, extending TNO's published pre-treatment assessments and mass-balance based recovery analysis. The battery passport module follows the EU requirements for interoperable, machine-readable, access-controlled data and builds on TNO's published reference architecture and demonstrator implementations. Finally, secure data exchange is enabled via an IDSA-certified connector (TNO Security Gateway) to share selected passport and processing artifacts (for example composition, state-of-health indicators, discharge certificate, and mass-balance results) under controlled access rights.

Proposed Concept:

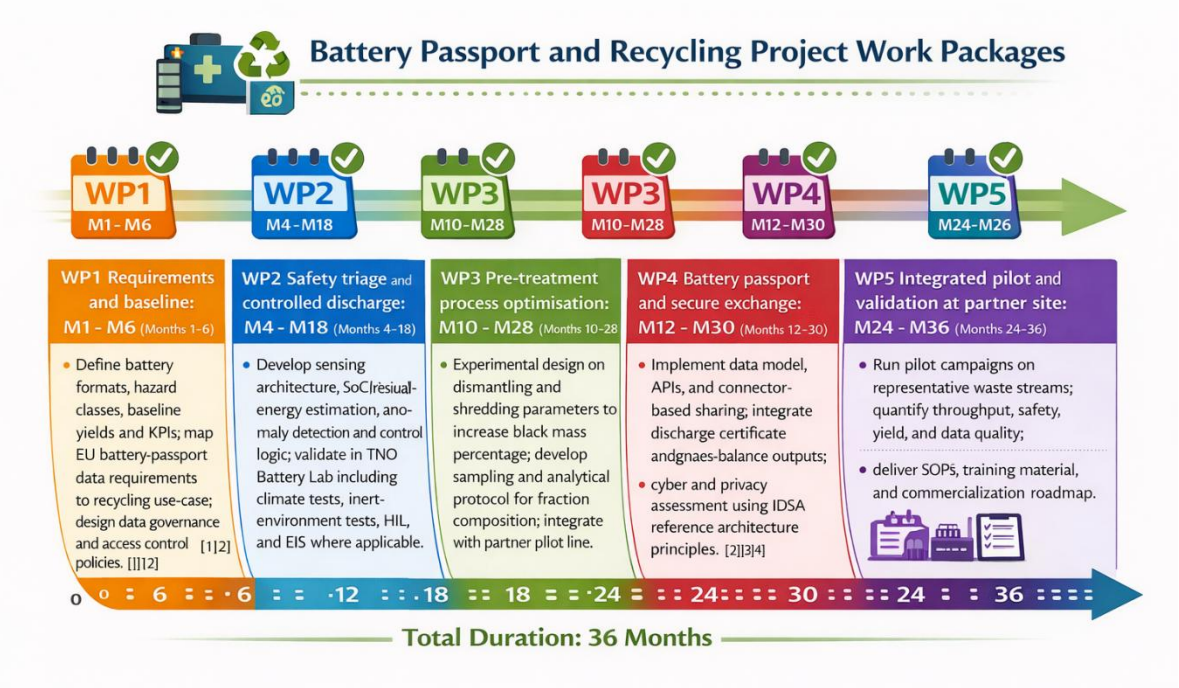
- Target stage: recycling pre-treatment (discharging, dismantling, shredding) plus a digital traceability layer that records process and battery condition data in a battery passport.
- Safety and efficiency module: risk-aware triage and controlled discharge of incoming EV packs/modules using electrical measurements (V, I), temperature sensing, and gas detection;

	optional Electrochemical Impedance Spectroscopy (EIS) and Hardware-in-the-Loop validation at TNO's facilities. [6][7] - Battery state modelling: remaining useful life and degradation models (including work co-authored by Steven Wilkins) to support decision logic for triage and routing to reuse vs recycling pathways. [9][10] - Process optimization module: parameter and equipment configuration for dismantling and shredding to maximize black mass yield and reduce losses of critical metals to coarse fractions, building on published TNO pre-treatment assessment results. [5] - Battery passport module: a data model and architecture aligned with EU battery passport requirements, based on TNO's battery-passport demonstrator and reference architecture work. [2][1] - Secure data exchange: controlled sharing of battery-passport data (e.g., composition, SoH, discharge certificate, mass balance) via an IDSA-certified data space connector (TNO Security Gateway). [3][4]
Rationale for Joint R&D Project	- The Korean SME ideally partner operates EV battery recycling and pre-processing (pack/module/cell dismantling and closed-loop recycling). The key bottlenecks for scale-up are safety, throughput, and consistent recovery yields. - TNO contributes validated methods and facilities for battery-system testing and algorithm validation, including climate-controlled testing from -40°C to +150°C, a 300 kW battery cycler, inert test environments with gas detection, and HiL and EIS-capable equipment. [6][7] - TNO also contributes a mature digital layer for battery passports and secure cross-organization data exchange, enabling compliance preparation for the EU battery passport that becomes mandatory from 2027 for relevant batteries. [1][2][3] - Joint R&D is required because the safety and process models must be calibrated on real industrial recycling lines and feedstock, while the digital passport requires interoperability with OEM and service-provider ecosystems.
Key Words	1. EV battery recycling pre-treatment 2. Risk-aware discharge and dismantling 3. Black mass yield optimisation 4. Battery passport (EU 2023/1542) 5. Secure data spaces (IDSA, TNO Security Gateway)

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2)]	Start of the Project:	Level (4)	End of the Project:	Level (6)
R&D Objectives	- Develop and validate a risk-aware triage and controlled discharge module for incoming EV battery packs/modules, including SoC and residual-energy estimation and anomaly detection suitable for recycling pre-treatment. - Develop a pre-treatment process model and control guidelines (dismantling, shredding, separation) to increase black mass yield and reduce critical-material losses to coarse fractions. - Implement a battery-passport compliant data pipeline that captures pre-treatment and quality data and supports controlled sharing via an IDSA-compatible connector (TNO Security Gateway). [1][2][3][4] - Demonstrate an integrated pilot on representative EV battery formats (NMC and LFP modules/packs) and deliver SOPs, safety case, and scale-up recommendations.			
R&D Target Specifications	- Safety: 0 safety incidents during pilot discharging and pre-treatment demonstrations; alarm and shutdown reaction time < 1 s for over-temperature or gas-detection triggers (validated in lab and partner pilot). - Discharge: reduce module terminal voltage to <= 30 V (and/or cell voltage to <= 2.0 V when applicable) within <= 60 minutes per module under controlled conditions; record time-current-temperature profile as a 'discharge certificate' stored in passport record. - Diagnostics: SoC estimation error <= 5% (absolute) during controlled discharge for representative module types; classification accuracy >= 90% for pre-defined hazard classes (normal, damaged, thermal-risk) using available sensor set. - Materials recovery: increase black mass yield (mass% of feed) by >= 5% versus partner baseline for selected NMC module design; document metal distribution across fractions via sampling and mass-balance. - Traceability: create a standardized, machine-readable record per processed pack/module including composition (if available), SoH indicators, pre-treatment parameters, and mass-balance; enable secure data sharing with external stakeholders via connector. [1][2][3]			

R&D Plan



Total duration: 36 months (suggested).

WP1 (M1-M6) Requirements and baseline: define battery formats, hazard classes, baseline yields and KPIs; map EU battery-passport data requirements to recycling use-case; design data governance and access control policies. [1][2]

WP2 (M4-M18) Safety triage and controlled discharge: develop sensing architecture, SoC/residual-energy estimation, anomaly detection and control logic; validate in TNO Battery Lab including climate tests, inert-environment tests, HiL, and EIS where applicable. [6][7]

WP3 (M10-M28) Pre-treatment process optimisation: experimental design on dismantling and shredding parameters to increase black mass percentage; develop sampling and analytical protocol for fraction composition; integrate with partner pilot line.

WP4 (M12-M30) Battery passport and secure exchange: implement data model, APIs, and connector-based sharing; integrate discharge certificate and mass-balance outputs; cyber and privacy assessment using IDSA reference architecture principles. [2][3][4]

WP5 (M24-M36) Integrated pilot and validation at partner site: run pilot campaigns on representative waste streams; quantify throughput, safety, yield, and data quality; deliver SOPs, training material, and commercialization roadmap.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	- System design and research leadership for safety diagnostics, controlled discharge strategies, and process modelling. - Develop model-based discharge strategies using TNO's Battery Modular Simulation Tool (BMST) for modular modelling and simulation of battery-system behaviour under discharge and thermal constraints. [8] - Provide access to TNO Battery Lab and test facilities for validation of sensors, algorithms and discharge protocols, including climate-controlled testing (-40°C to +150°C), inert test environments with gas detection, and 300 kW battery cycler. [6][7] - Develop the battery-passport data model and demonstrator components (data pipeline, access control) building on TNO battery passport architecture and demonstrator work. [2] - Implement secure data-sharing connector layer based on the TNO Security Gateway and support integration with partner IT/OT systems. [3][4] - Deliver: validated algorithms, prototype software/hardware specifications, pilot-test reports, and standard operating procedures for scale-up.
Expected R&R of Korean SME Partner	- Provide representative end-of-life EV battery feedstock (modules/packs) and baseline process data for pre-treatment (dismantling, discharging, shredding and separation) under appropriate safety procedures. - Host and operate the pilot-scale pre-treatment line for validation (industrial environment), including implementation of discharge workstation and data-capture interfaces. - Support mechanical design adaptations and integration of sensors/controls in the pre-treatment equipment; provide operators for trial campaigns and feedback for SOP development. - Perform and co-design fraction sampling and material quality measurements (black mass, coarse fraction) and support techno-economic assessment for scale-up and commercialization.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	- Improved safety and throughput of EV battery pre-treatment via validated triage and controlled discharge protocols, reducing handling risk of waste Li-ion batteries. [6][7] - Higher and more consistent black mass yield through process improvements that reduce losses of valuable metals to coarse fractions. [5] - Operational battery-passport integration for recycling, creating a standardized record of discharge, pre-treatment parameters and mass-balance that supports circular value creation and prepares for EU battery passport obligations. [1][2] - Secure and scalable data exchange between recycler, OEMs and service providers via an IDSA-compatible connector (TNO Security Gateway), enabling controlled sharing of sensitive battery data. [3][4] - Commercialization potential: a pre-treatment safety and data-traceability package that can be deployed in recycling plants to differentiate services and enable premium closed-loop contracts.

6. References

[1] EUR-Lex. Regulation (EU) 2023/1542 of the European Parliament and of the Council of 12 July 2023 concerning batteries and waste batteries (Text with EEA relevance). OJ L 191, 28.7.2023. URL: https://eur-lex.europa.eu/eli/reg/2023/1542/oj/eng (accessed 2026-01-24).

[2] Deventer, M.O., et al. TNO. Towards future-proof battery passports. Report No. TNO 2024 R10003, 2024. URL: https://publications.tno.nl/publication/34643201/Ky5iElkO/TNO-2024-R10003.pdf (accessed 2026-01-24).

[3] TNO. TNO Security Gateway certified for data spaces (IDSA certification). News item, 13 March 2024. URL: https://www.tno.nl/en/newsroom/2024/03/tno-security-gateway-certified-data-spaces/ (accessed 2026-01-24).

[4] TNO. TNO Security Gateway (TSG) Documentation (IDS connector). URL: https://tno-tsg.gitlab.io/tsg-documentation/ (accessed 2026-01-24).

[5] Beek-Hagemans, J., et al. TNO. D6.3 Li-ion battery pretreatment process assessment (Green Transport Delta - Electrification). 5 September 2024. URL: https://publications.tno.nl/publication/34643112/CNRwUE9X/beek-hagemans-2024-li-ion.pdf (accessed 2026-01-24).

[6] TNO. Sustainable batteries with advanced battery management (incl. Battery Lab test capabilities). Web page. URL: https://www.tno.nl/en/focus-areas/mobility-logistics/roadmaps/zero-emission-mobility/batteries/sustainable-batteries-advanced-battery-management/ (accessed 2026-01-24).

[7] European Electric Mobility Center (Automotive Campus). Vehicle and battery testing at the European Electric Mobility Center. PDF, 2022. URL: https://www.automotivecampus.com/wp-content/uploads/vehicle-and-battery-testing-at-the-european-electric-mobility-center-2022.pdf (accessed 2026-01-24).

[8] Dadikozyan, A., Hoedemaekers, E., Hoekstra, F., Wilkins, S. Modular modelling and simulation environment for battery design and analysis - Battery Modular Simulation Tool (BMST). EVS37 Symposium, 23-26 April 2024. URL: https://www.researchgate.net/publication/380185719_Modular_Modelling_and_Simulation_Environment_for_Battery_Design_and_Analysis_-_Battery_Modular_Simulation_Tool_BMST_- (accessed 2026-01-24).

[9] Geerts, D., Medina, R., van Sark, W., Wilkins, S. Charge Scheduling of Electric Vehicle Fleets: Maximizing Battery Remaining Useful Life Using Machine Learning Models. Batteries (MDPI), 2024. URL (TNO repository PDF): https://publications.tno.nl/publication/34642199/vW8uDg/geerts-2024-charge.pdf (accessed 2026-01-24).

[10] Wilkins, S. Battery Materials for NL: Towards a Circular Battery Economy. MaterialenNL 2023 (conference), December 2023. URL: https://www.researchgate.net/publication/376751897_Battery_Materials_for_NL_Towards_a_Circular_Battery_Economy (accessed 2026-01-24).

[11] TNO. Battery Lab (facility overview). Web page. URL: https://www.tno.nl/en/technology-science/facilities/battery-lab/ (accessed 2026-01-24).

[12] TNO. Recovering critical raw materials from LFP batteries. News item, 21 January 2025. URL: https://www.tno.nl/en/newsroom/2025/01/recovering-raw-materials-batteries/ (accessed 2026-01-24).

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization